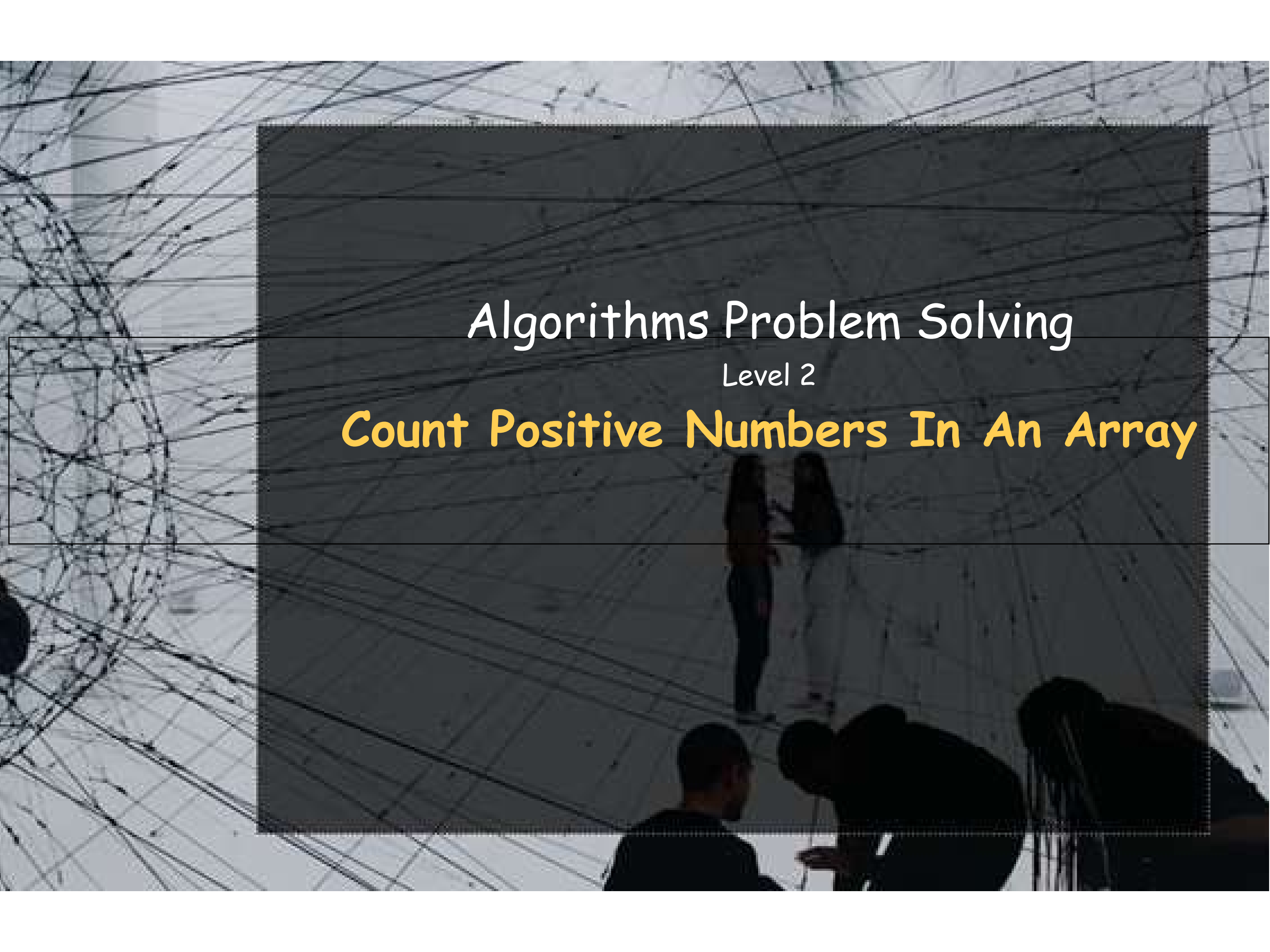


Algorithms Problem Solving

Level 2

Count Positive Numbers In An Array



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Definition

- ◆ This program fills an array with random numbers from ``-100`` to ``100``.
- ◆ It counts how many positive numbers exist in the array.
- ◆ Finally, it prints the array elements and the total count of positive numbers.

Explanation

- ◆ `FillArrayWithRandomNumber()` fills the array with random numbers between `-100` and `100`.
- ◆ `CountPositiveNumbers()` loops through the array and checks each number.
- ◆ If the number is greater than or equal to `0`, `count` is increased by `1`.
- ◆ The condition used in the code is:

```
if (arr[i] >= 0)
    count++;
```
- ◆ This means that `0` is counted as a positive number in this program.
- ◆ Mathematically, `0` is neither positive nor negative.
- ◆ If we want to count strictly positive numbers, the condition should be:

```
if (arr[i] > 0)
    count++;
```

Steps in main()

1. Ask the user for the number of elements.
2. Fill the array with random numbers from `-100` to `100`.
3. Print all array elements.
4. Loop through the array.
5. Check whether each number is greater than or equal to `0`.
6. Increase `count` when the condition is true.
7. Return the total count.
8. Print the number of positive numbers.

💡 Example

◆ `Array Elements` :

-32 -46 -2 71 53 -20 13 36 30 48

◆ Numbers counted by the current code:

71 53 13 36 30 48

◆ Therefore, the current code counts `6` numbers, not `5`.

◆ The example says `5`, but the provided array actually contains six positive numbers.

➡ Result

Array Elements: -32 -46 -2 71 53 -20 13 36 30 48

Positive Numbers count is : 6

Note

- ◆ The function is named `CountPositiveNumbers()` , but the condition `arr[i] >= 0` also includes `0` .
- ◆ For a mathematically correct definition of positive numbers, use:
 if (arr[i] > 0)
 count++;
- ◆ `count` starts from `0` and increases whenever a qualifying number is found.
- ◆ `ReadArrayElements()` and `AddElementToArray()` are defined but not used in this program.

⊘ Warning

- ◆ `arr[100]` can store a maximum of `100` elements, so `arrLength` should not be greater than `100`.
- ◆ `FillArrayWithRandomNumber()` does not validate the entered length.
- ◆ `cmath` and `corecrt_math_defines.h` are not needed.
- ◆ `rand()` and `time()` should ideally have their required headers included explicitly.
- ◆ The example output says `5`, but based on the provided numbers, the correct count is `6`.
- ◆ If `0` appears, the current code counts it as positive.

Comparison

▨ ``arr[i] > 0``

- ✓ Counts strictly positive numbers.
- ✓ Does not count ``0``.
- ✓ Mathematically correct for positive numbers.

▤ ``arr[i] >= 0``

- ✓ Counts positive numbers and ``0``.
- ✓ Does not count negative numbers.
- ✓ This is the condition used in the current program.

✦✦ Summary

- ✓ Fill the array with random numbers from `-100` to `100`.
- ✓ Check every element.
- ✓ Count numbers that satisfy the condition.
- ✓ The current code uses `>= 0`, so `0` is included.
- ✓ The provided example contains `6` positive numbers, not `5`.

Thank you 😊